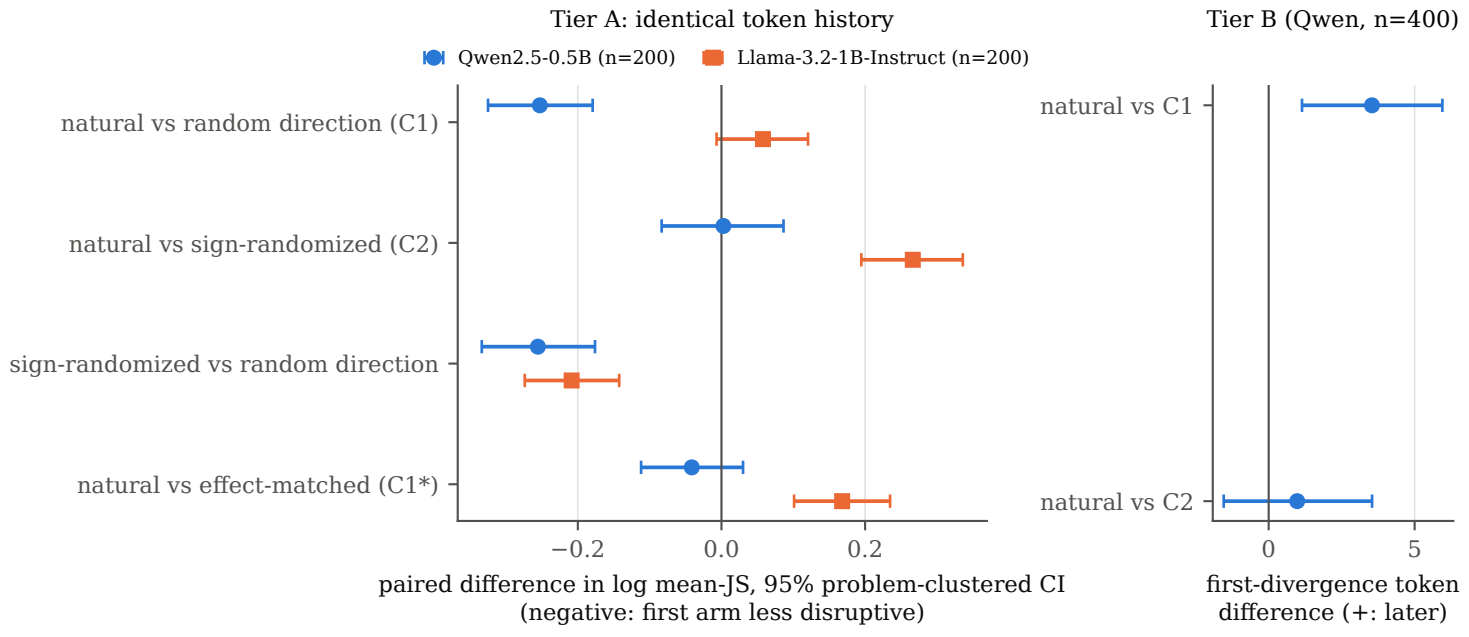
Tier A: identical token history
Tier B (Qwen, n=400)
Qwen2.5-0.5B (n=200)
Llama-3.2-1B-Instruct (n=200)
natural vs random direction (C1)
natural vs sign-randomized (C2)
sign-randomized vs random direction
natural vs effect-matched (C1*)
−0.2
0.0
0.2
paired difference in log mean-JS, 95% problem-clustered CI
(negative: first arm less disruptive)
natural vs C1
natural vs C2
0
5
first-divergence token
difference (+: later)